

Answer Key

Supplementary Problem 2: System of Particles

1 System Data & Total Mass

Given Particle Coordinates and Masses

The system consists of five particles distributed in a 2D Cartesian plane:

- $m_1 = 3 \text{ kg}$ at $A(2, 2)$
- $m_2 = 1 \text{ kg}$ at $B(-2, 1)$
- $m_3 = 2.5 \text{ kg}$ at $C(-1, -3)$
- $m_4 = 1.5 \text{ kg}$ at $D(3, -1)$
- $m_5 = 2 \text{ kg}$ at $E(0, 4)$

Calculation of Total Mass (M):

$$M = \sum_{i=1}^5 m_i = 3 + 1 + 2.5 + 1.5 + 2$$

$$M = 10 \text{ kg}$$

2 Abscissa of Center of Mass (x_G)

Using the weighted average formula for the x -coordinates:

$$x_G = \frac{\sum m_i x_i}{M} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4 + m_5 x_5}{10}$$

Substitution:

$$x_G = \frac{3(2) + 1(-2) + 2.5(-1) + 1.5(3) + 2(0)}{10}$$

Simplification:

$$x_G = \frac{6 - 2 - 2.5 + 4.5 + 0}{10} = \frac{6}{10}$$

$$x_G = 0.6 \text{ m}$$

3 Ordinate of Center of Mass (y_G)

Using the weighted average formula for the y -coordinates:

$$y_G = \frac{\sum m_i y_i}{M} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3 + m_4 y_4 + m_5 y_5}{10}$$

Substitution:

$$y_G = \frac{3(2) + 1(1) + 2.5(-3) + 1.5(-1) + 2(4)}{10}$$

Simplification:

$$y_G = \frac{6 + 1 - 7.5 - 1.5 + 8}{10} = \frac{6}{10}$$

$$y_G = 0.6 \text{ m}$$

4 System Visualization

The Center of Mass for this system is located at point $G(0.6, 0.6)$.

Visual Rationale

Point G is situated in the first quadrant, very close to the origin. This reflects the balance between the heavy mass at $A(3\text{kg})$ and $E(2\text{kg})$ against the negative coordinates of B and C .

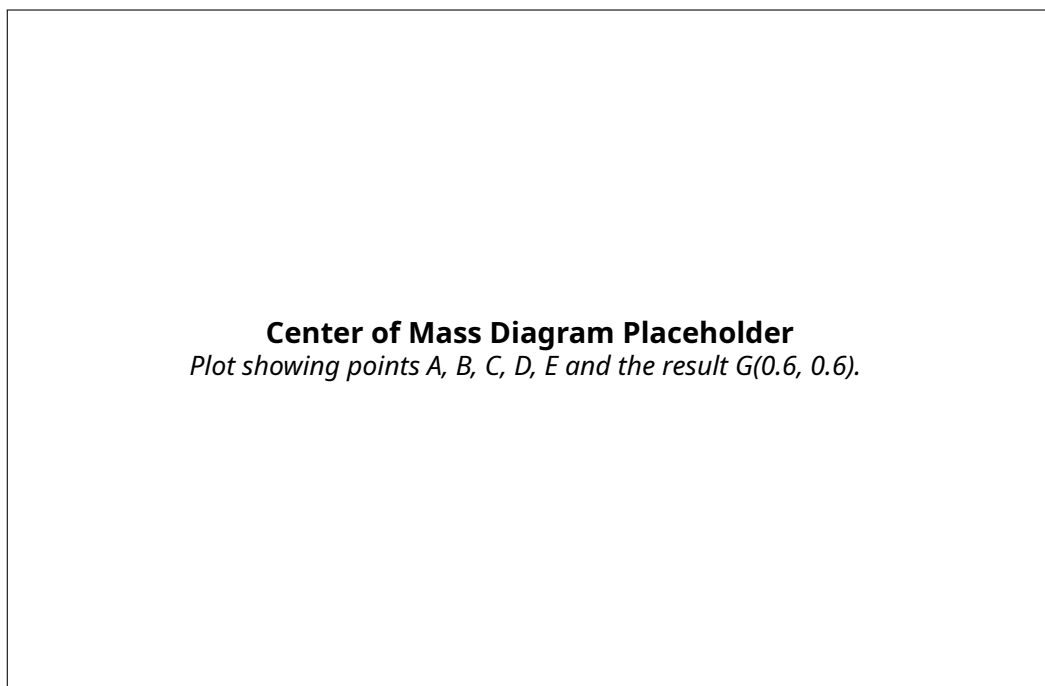


Figure 1: Spatial distribution of the 5-particle system and the resulting Center of Mass (G).

End of Worked Solution